# Smart Emergency Response System for Road Accidents with Automatic Accident Detection

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## Abstract

Almost four hundred people lose their lives per day in road accidents in Indian subcontinent alone [3]. We know that the first few minutes play a vital role in deciding the fate of the victim involved in the accident. The Emergency Response Support System (ERSS) around the world are now improving their infrastructure to find the accurate location of the distress call. On the other hand, sometimes it is not possible for the victim to call the emergency services. Till now, most of the accident detection and response mechanisms focused on using specialised hardware solutions to detect the crash with a few exceptions.
This paper presents a way of detecting road accidents through a smartphone that avoids false positives. An Android smartphone-based prototype architecture of our accident detection system along with empirical results are being presented.

## 1 Introduction

It has been almost a decade since the beginning of smartphone era that started by launch of iPhone. It has been one of the most empowering inventions that mankind has ever seen. By 2016 around 2.1 billion smartphones were active and during 2018 that number has grown to 5.135 billion that helped in pulling 4.021 billion people online [5][1]. Smartphone is the most advanced package of sensors thats available to almost two- thirds of global population. Nearly 1.25 million people die in road crashes each year, on average 3,287 deaths a day in India [3]. While this might not be a big media event, but it is more than the number of causalities in air crash or mass shootings. A considerable number of people die due to slow response to the emergency situation [19]. Automated car accident detection can save lives by decreasing the time required for information to reach emergency responders.
This research paper provides making emergency services accessible to the victim automatically and making victims information available to emergency respondent while informing family members and the institution from which the person is related to about the emergency.
Developing a comprehensive emergency response system for road accidents can help in reducing fatalities caused due to the same significantly. But to accelerate process of adoption of such system, the hardware needs to be portable and mass -adopted which at present is smartphone. Smartphone with its on-board sensors is capable of detecting accelerations that are higher than that experienced in day-to-day activities even while they might not detect accelerations that are experienced during serious accidents. This paper finds a threshold acceleration that is seldom experienced in day-to-day life while it is definitely experienced during the road accidents. We also describe how our threshold is designed to prevent false positives and how this system can be used to provide analytics data to government agencies for making policies and infrastructure keeping safety in mind.

This paper contributed the following

- First, solutions to key challenges associated with detecting traffic accidents, such as preventing false positives.
- Second, a porotype architecture of smartphone-based accident detection system and empirical analyse and its ability to resist false positives as well as its capabilities for accident reconstruction.
- Third, discuss on the process of smartphone-based accident detection that reduce overall traffic congestion and increase the preparedness of emergency responders.

## 2 Related Works

Detecting accidents automatically using a dedicated device has been interest of academia since a long time. Proposed solutions for using smartphones for detecting and communicating road accidents have existed ever since smartphones came in mass market. Iyyappan et al. [14] presented an approach to detect and communicate accidents using a dedicated device fitted with relevant sensors and microprocessors. Ali et al. [9] in another paper used a dedicated device for accident detection but relayed the data to smartphone for communication of the accident. In 2011,Thompson et al. came up with WreckWatch [21], an application that detected accidents using smartphones and provided situational awareness to the emergency respondents. They used acceleration filter and microphone filter to detect the road accidents. However, amid privacy and security concerns using microphone filter in background is not such a

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good idea. Thus, our paper stresses on using only acceleration filter to detect road accidents. The acceleration filter is high enough to be difficult to produce manually and low enough to detect accident. In rare event of false detection of accident, an alarm starts ringing which gives one minute to user to cross out the detection of accident and if user does not respond for one minute it is assumed that accident has happened.

## 3 Methodology

Android today is used in 84.8 % of smartphones globally [8]. That made android the first choice of deployment for the Emergency Response System for Road Accidents. Almost, all the android phones have five major sensors Microphone, Camera, GPS/GLONASS, Proximity Sensor and 3-axis Accelerometer with sensitivity of 16G [2]. Majorly only two phenomena were of interest to classify an event into a serious accident. First one was to look for unusual spikes in acceleration and the second thing to look out was to analyse the sound produced just after the first event. But amidst privacy concerns [15] the given system is not using the second factor. However, even in absence of microphone data, accelerometer can predict the accident event nearly as accurate.

### 3.1 Android Client

The features discussed above have been integrated in an Android App [17] called **Safely** (as shown in Figure 1) which is designed to both satisfy the purpose of detecting accident accurately and feel intuitive to user. It starts with password less login using Facebook Account Kit [18]. Then there is a small signup form (Figure 2) which takes critical health and profile information and sends them to a secure Firebase Server [16]. That opens up a start emergency screen where software button can be pushed to activate emergency situation (as shwon in 4). Login event also triggers an activity that triggers emergency situation by long pressing power key. User can go to settings and manually turn on Automatic Accident Detection to activate automatic accident detection module in background.

### 3.2 Challenges in Detecting Accidents Automatically

Modern vehicles come equipped with ECUs that have on-board sensors and capabilities to detect the accidents [13]. But the most difficult challenge comes in developing a scalable solution that can interface with a wide variety of ECUs. It also hardly makes any sense economically to make a dedicated hardware for detecting accidents automatically. So, smartphones will hardly be able to detect certain metrics about number of times a vehicle is hit, data about airbag deployment, angle of impact etc. Also, it unrealistic for a user to connect their smartphones to ECUs before every drive. That makes it almost impossible to get significant data from vehicles onboard Event Data Recorders [4] and that problem is magnified even further due to the lack of common interface for connecting to EDRs and sometimes lack of EDRs in some cars. So, using smartphones sensors to detect high accelerations and decelerations




Figure 1. Sign In/Up Screen .

can help us to detect accidents. But even crucial is to develop a system that does not work with false positives. Phones are subject to fall and heavy braking should not be detected as an accident.

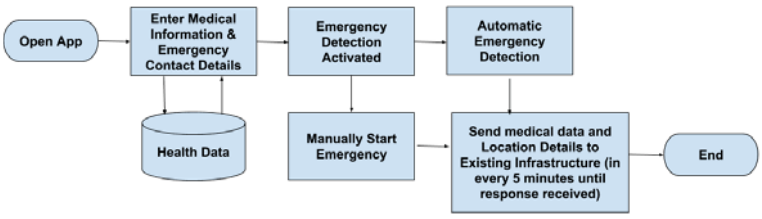



Figure 6. Flowchart of events once emergency situation is triggered

### 3.3 Hardware Used

All experiments were performed on HMD Globals Nokia 7 Plus device. It comes with Snapdragon 660 processor coupled with 4 GB RAM [6]. It comes equipped with Proximity Sensor, Accelerometer, Gyroscope, Microphone and Magnetometer. Its accelerometer can detect accelerations upto 16G.

### 3.4 Solution Approach

In modern world theres a global trend of increasing smartphone penetration [7], a growing number of people are going to have an array of sensors in their pocket. These sensors essentially empower us to detect a lot of patterns. It is also possible to predict the accidents automatically using the same. However, there are many challenges associated with using smartphone to detect accidents. There are lot of cars that already come within built accident detection and emergency response systems [12]. They are specifically designed for this case and predict accidents with high accuracy. However, it is limited to less than ten percent of total vehicles. But smartphones are used in various other circumstances and are likely to raise false alarms. Thus,

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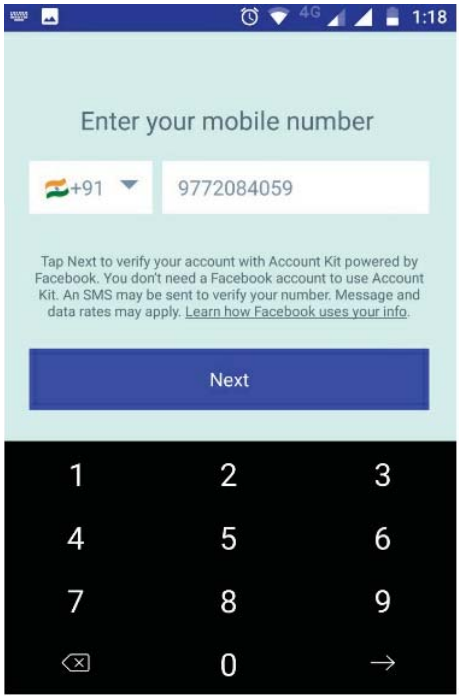



Figure 2. Enter Mobile Number

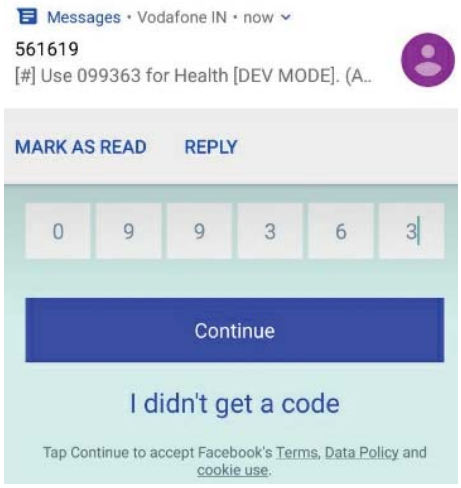



Figure 3. Enter OTP

eliminating false positives due to situations like dropping the phone is of prime importance. Also, the orientation of phone might affect the accident detection. Of course, the interaction with the Electronic Control Unit frequently is quite tiresome and not feasible. There is also a need provide the responder with situational awareness and criticality of the accident to the responders.

We need to develop a comprehensive system despite these challenges and a universal challenge of conserving the battery as much as possible.

In this case, user has to activate automatic accident detection module in app before starting the drive. When an accident occurs, phone being in the vehicle experiences same forces as vehicle itself [10]. Then if at any instant the acceleration goes above 13G [20], accident is detected, and a message is sent to the emergency contact selected by the user during sign in and a contact from users workplace along with the users live location.

### 3.4.1 Impact during the accident

Force of impact is the force generated when objects meet. So, its the force that vehicles experience during accidents. It can be derived using law of conservation of energy. If a car is moving, then it possesses kinetic energy that reduces to zero after the accident. Thus, change in kinetic energy must be compensated by the work done by the impact force to fulfil the law of conservation of energy.

Hence,

$$F = m * \frac{v^2}{2 * d} \tag{1}$$

where,
F = average impact force
m = mass of the object
v = initial speed of the object
d = distance travelled during collision

In case of accident, extending the distance moved during the collision drastically reduces the average impact force. Alternatively, we can write the above equation as:

$$F = \frac{v * m}{t} \tag{2}$$

Thus, from above equation we can conclude that extending the time of collision decreases the average impact force. This can be easily visualized by considering an example where a man jump from a certain height to the ground and to a trampoline. In case of trampoline the man experiences less average impact force due to extended time of collision. This is analogical to vehicular crashes where extending the time and/or distance reduces the impact as a result of crash. In fact, extending the time of collision is the main principle of working of seat belts. So, to understand this impact force let us consider Table 1 given below to develop an intuition for understanding the following Impact Force vs Speed graph as shown in Figure 9.

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| S.no. | Total Weight of Passengers (Kg) | Car Speed (Km/h) | Average Impact force (kN) | Stopping Time (ms) | Deceleration (g) |
|---|---|---|---|---|---|
| 1 | 180 | 30 | 31.25 | 48 | 18 |
| 2 | 180 | 50 | 86.8 | 29 | 49 |
| 3 | 180 | 70 | 170 | 20.6 | 96 |
| 4 | 180 | 90 | 281 | 16 | 159 |
| 5 | 180 | 110 | 420 | 13 | 238 |
| 6 | 180 | 130 | 587 | 11 | 332 |
| 7 | 180 | 150 | 781 | 9.6 | 443 |

Table 1. Variation in Average impact forces w.r.t. Speed




Figure 4. Landing page with virtual button

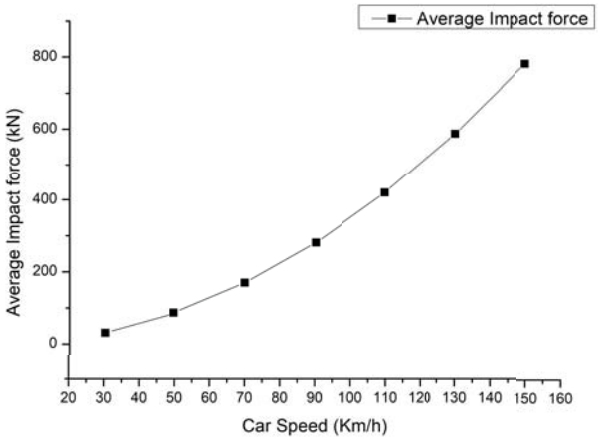



Figure 7. Impact force vs Speed

### 3.5 Eliminating False Positives

The major goal of this application was to eliminate false positives due to heavy breaking or fall and conserve battery. Given

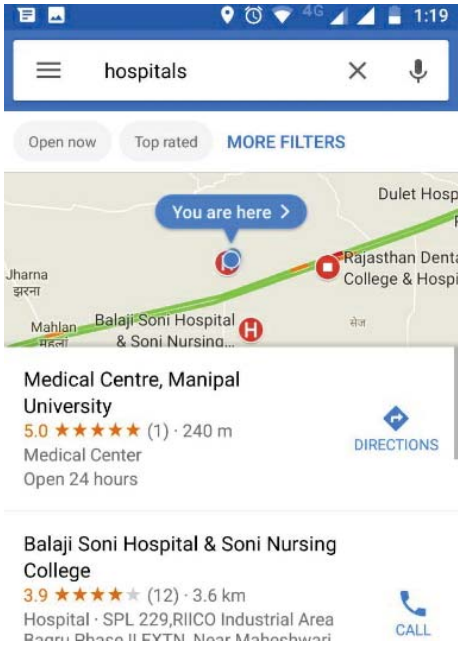



Figure 5. Nearest Hospitals

below are the filters used to eliminate false positives:

- GPS consumes a substantial power. Therefore, the automatic accident detection module is required to be activated manually to run in background.
- Normal phones can detect acceleration upto 16Gs. During free fall accelerometer cannot detect acceleration of more than 3Gs and even throwing the phone hard will make the phone experience a maximum acceleration of just 9Gs. While some of the airbags deploy at accelerations of upto 60G [11], 13G could be a safe operational limit to detect accidents. While, airbags experience the same acceleration/deacceleration as that of vehicle itself, riders inside the car might experience much lower acceleration/deacceleration due to seat belts, thus, making mobile phone to experience less acceleration. Hence, 13G could be a safe operational limit to detect accidents which is much lower than 60G used in airbags.
- Heavy braking from 105 km/hr to 0 km/hr also does not produce acceleration of more than 10G for the person having mobile phone in the pocket.

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## 4 Empirical Results

Avoiding false positives is a major challenge when detecting car accidents with mobile devices. To study the potential for false positives, we did two demo experiments designed to simulate environment that trigger accelerations whose values could be interpreted as car accidents. For the first test, the mobile device was dropped from ear height in the drivers seat of a car. Another scenario that could potentially create a false positive is a sudden stop. This test was performed in a car by reaching a speed of approximately 105 kmph and engaging in a sudden stop. The test results are approximate as the exact speed was unknown and braking pressure was also not exactly known. Figure 8 below shows the acceleration experienced on each axis during the stop. As described above, because the smartphone remained stationary relative to the vehicle, it experienced the same forces as the vehicle. In this instance, the acceleration experienced by the smartphone was less than that experienced during the fall. This result is attributed to the fact that although the stop was sudden and forceful, the car (and consequently the smartphone) came to a rest over a duration of time that was longer than during the drop test. In other words, the change in velocity was greater but the actual acceleration was less because the change occurred over a longer duration of time.

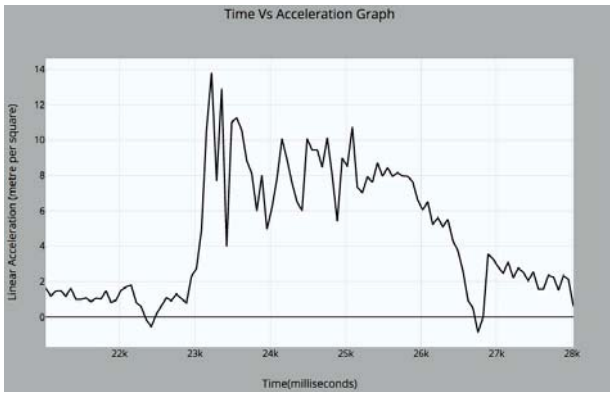



Figure 8. Time vs Acceleration plot during heavy braking

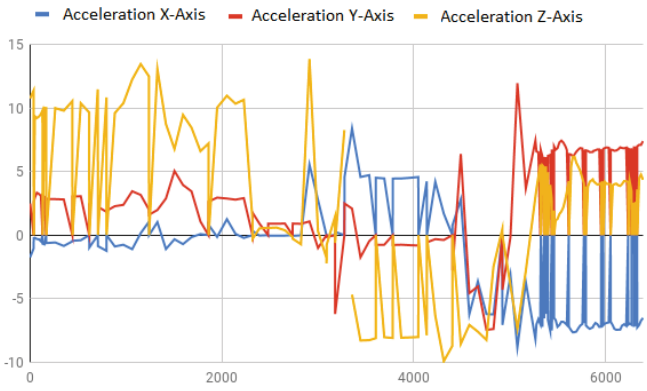



Figure 9. Acceleration plot during sudden fall

## 5 Conclusion

In this paper, we propose a novel way of detecting accidents that does not use microphone in background and detects the accident using accelerometer of the smartphone. The goal of this research was to develop an Emergency Response System for Road Accidents thats portable, accessible and avoids false positives. This paper also describes how filters are used to effectively remove false positives and how force generated even in very heavy braking is less than the filter. However, this system does suffer from limitations especially if the phone is damaged in road accidents. But nevertheless, the benefits of such a system go beyond just detecting road accidents. The data from accidents such as location of accident, age group of victims, timings of accidents etc can be of extensive use to the governments for improving roads and infrastructure that are safer for the public.

## References

[1] Global digital 2018 report. https://wearesocial.com/us/blog/2018/01/global-digital-report-2018. Accessed on: 2019-02-20.

[2] Report by android.com. https://source.android.com/devices/sensors/sensor-types. Accessed on: 2019-02-20.

[3] Report by bbc on road accidents in india,2015. https://www.bbc.com/news/world-asia-india-36496375. Published on: 2016-06-10.

[4] Report by national highway traffic safety administration, united states department of transportation. https://www.nhtsa.gov/research-data/event-data-recorder. Accessed on: 2019-02-20.

[5] Report by statista.com on number of smartphone users worldwide from 2014 to 2020 (in billions). https://www.statista.com/statistics/330695/number-of-smartphone-users-worldwide/. Accessed on: 2019-02-20.

[6] Report on device specification nokia 7 plus. https://www.nokia.com/phones/en$_{i}nt/nokia - 7 - plus. Accessedon$ : 2019 − 02 − 20.

[7] Report on smartphone sales statics. https://www.statista.com/topics/840/smartphones/. Accessed on: 2019-02-20.

[8] Smartphone market share- a report by idc. https://www.idc.com/promo/smartphone-market-share/os. Accessed on: 2019-02-20.

[9] Asad Ali and Mohamad Eid. An automated system for accident detection. In *2015 IEEE International Instrumentation and Measurement Technology Conference (I2MTC) Proceedings*, pages 1608–1612. IEEE, 2015.

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[10] Haimd M Ali and Zainab S Alwan. *Car accident detection and notification system using smartphone*. LAP LAMBERT Academic Publishing Saarbrucken, 2017.

[11] Jo Barnes, Andrew Morris, Brian Fildes, and Stuart V Newstead. Airbag effectiveness in real world crashes. 2001.

[12] Manuel Fogue, Piedad Garrido, Francisco J Martinez, Juan-Carlos Cano, Carlos T Calafate, and Pietro Manzoni. Automatic accident detection: Assistance through communication technologies and vehicles. *IEEE Vehicular Technology Magazine*, 7(3):90–100, 2012.

[13] Varsha Goud. Vehicle accident automatic detection and remote alarm device. *International Journal of Reconfigurable and Embedded Systems*, 1(2):49, 2012.

[14] Mr S Iyyappan and Mr V Nandagopal. Automatic accident detection and ambulance rescue with intelligent traffic light system. *International journal of advanced research in electrical, electronics and instrumentation engineering*, 2(4):1319, 2013.

[15] Asma Khatoon and Peter Corcoran. Privacy concerns on android devices. In *2017 IEEE International Conference on Consumer Electronics (ICCE)*, pages 149–152. IEEE, 2017.

[16] Chunnu Khawas and Pritam Shah. Application of firebase in android app development-a study. *International Journal of Computer Applications*, 975:8887, 2018.

[17] Jianye Liu and Jiankun Yu. Research on development of android applications. In *2011 4th International Conference on Intelligent Networks and Intelligent Systems*, pages 69–72. IEEE, 2011.

[18] Michitomo Morii, Hiroki Tanioka, Kenji Ohira, Masahiko Sano, Yosuke Seki, Kenji Matsuura, and Tetsushi Ueta. Research on integrated authentication using passwordless authentication method. In *2017 IEEE 41st Annual Computer Software and Applications Conference (COMPSAC)*, volume 1, pages 682–685. IEEE, 2017.

[19] Stefan Rauscher, Georg Messner, Peter Baur, Jeffrey Augenstein, Kennerly Digges, Elana Perdeck, George Bahouth, and Oliver Pieske. Enhanced automatic collision notification system-improved rescue care due to injury prediction-first field experience. In *The 21st International Technical Conference on the Enhanced Safety of Vehicles Conference (ESV)-International Congress Center Stuttgart, Germany*, pages 09–0049, 2009.

[20] Dennis F Shanahan. Human tolerance and crash survivability. *Pathological Aspects and Associate Biodynamics in Aircraft Accident Investigation*, 2004.

[21] Jules White, Chris Thompson, Hamilton Turner, Brian Dougherty, and Douglas C Schmidt. Wreckwatch: Automatic traffic accident detection and notification with smartphones. *Mobile Networks and Applications*, 16(3):285–303, 2011.

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